Class Activity#4

1)Add

1. Example

```
MOV AL, 10H      ; Move 10H (16 decimal) to AL (8-bit register)
MOV BL, 20H      ; Move 20H (32 decimal) to BL
ADD AL, BL       ; AL = AL + BL = 30H (AL now contains 30H)
```

2. Example

```
MOV AX, 1000H    ; Move 1000H to AX (16-bit register)
MOV BX, 2000H    ; Move 2000H to BX
ADD AX, BX       ; AX = AX + BX = 3000H
```

2)Sub

1. Example

```
MOV AL, 50H      ; AL = 50H
MOV BL, 20H      ; BL = 20H
SUB AL, BL       ; AL = 50H - 20H = 30H (No borrow/Carry)
```

2. Example

```
MOV AX, 5        ; AX = 5
MOV BX, 10       ; BX = 10
SUB AX, BX       ; AX = 5 - 10 = -5 (in two's complement, this is 0xFFFB)
```

3)Increment

1. Example

```
MOV AL, 0        ; Start AL at 0
INC AL           ; AL = 1

MOV BL, 0FH      ; Set BL to 15 (0F Hex)
INC BL           ; BL = 16 (10 Hex)
```

2. Example

```
MOV CL, 0FFH     ; CL = 255 (Maximum value for 8-bit)
INC CL           ; CL overflows to 0 (because FF + 1 = 00 in 8-bit)
```

4)Decrement

1. Example

```
MOV BL, 5        ; Set BL to 5
DEC BL           ; BL becomes 4
```

2. Example

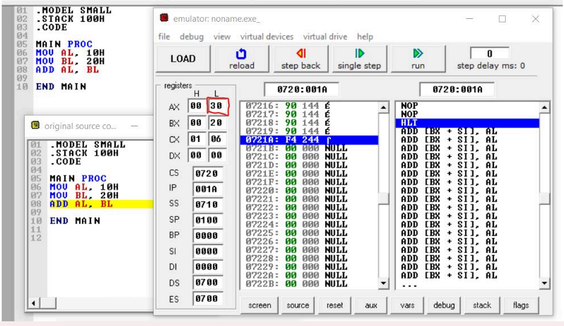
```
MOV [0500H], 5   ; Store 5 in memory
DEC [0500H]      ; Decrement value at memory to 4
```

Sample

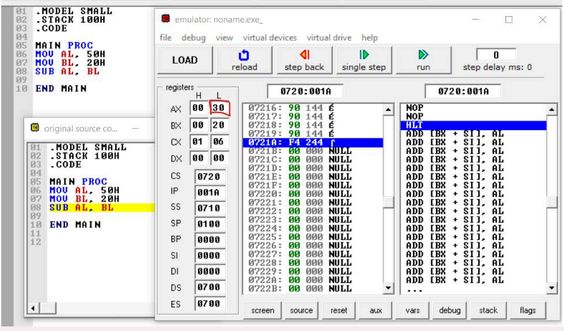
```
. MODEL SMALL

. STACK 100H

. CODE

    1)  ADD
    2)  SUB
    3)  INCREMENT
    4)  DECREMENT

END MAIN
```

```
.MODEL SMALL
.STACK 100H
.CODE

; 1> Add
MAIN PROC
MOV AL, 0
MOV BL, 0FH

END MAIN
```

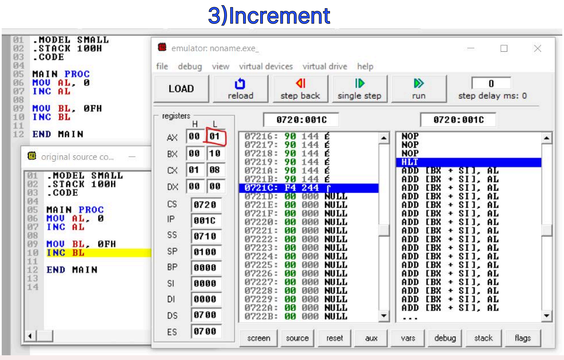
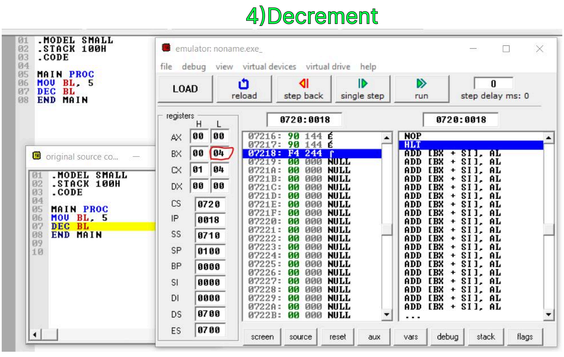
1)Add



2)Sub



3)Increment



4)Decrement